Neil Dey

843-245-4045 | neild799@gmail.com | neil-dey.github.io

128 State St S. Apt 341, Kirkland, WA, 98033

Education

North Carolina State University **GPA: 4.0**
PhD in Statistics *May 2025*

- Dissertation: *Valid Inference in Risk Minimization Problems*
- Advisors: Jonathan P. Williams & Ryan Martin

North Carolina State University **GPA: 4.0**
B.S. in Computer Science & Mathematics (with Honors) *2017 – 2020*

Peer-Reviewed Publications

Machine Learning Theory

- **Neil Dey** and Jonathan P. Williams. Valid Inference for Machine Learning Model Parameters. *R&R: Electronic Journal of Statistics*, 2024+.
  - Investigates how one can assign confidence levels to the location of the “optimal" parameter/hyperparameter vector of a machine learning model.
- **Neil Dey**, Ryan Martin, and Jonathan P. Williams. Generalized Universal Inference on Risk Minimizers. *Journal of the Royal Statistical Society Series B*, October 2025.
  - Extends the theory of the previous paper into the “online" setting (i.e. under both optional stopping and optional continuation of data collection).
- **Neil Dey**, Ryan Martin, and Jonathan P. Williams. Multiple Testing in Generalized Universal Inference. *Statistics & Probability Letters*, 2026
  - Further extends the previous papers to the setting where there may be multiple dependent data sources and multiple models to perform inference on.
- **Neil Dey**, Ryan Martin, and Jonathan P. Williams. Neil Dey, Ryan Martin, and Jonathan P Williams' contribution to the Discussion of “Safe Testing" by Grünwald, de Heide, and Koolen. *Journal of the Royal Statistical Society Series B*, June 2024.
  - Communicates the importance of various ideas discussed in the above papers to the general statistics community.

Natural Language Processing

- **Neil Dey**, Jing Ding, Jack Ferrell, Carolina Kapper, Maxwell Lovig, Emiliano Planchon, and Jonathan P. Williams. Conformal Prediction for Text Infilling and Part-of-Speech Prediction. *New England Journal of Statistics in Data Science*, October 2022.
  - Investigates how conformal prediction layers can be appended to existing NLP models (e.g., BERT) to produce set-valued predictions in downstream tasks, with the enhanced models guaranteeing correctly reported confidences in their predictions.
- **Neil Dey**, Matthew D. Singer, Srijan Sengupta, and Jonathan P. Williams. Word Embeddings as Statistical Estimators. *Shankya B*, May 2024.
  - Proposes a copula-based statistical model to synthetically generate natural language data; this model allows one to theoretically prove certain properties of static embedding models such as Word2Vec and GloVe.

Applications of Stochastic Processes

- **Neil Dey**, Adam Ousherovitch, Fiona Romanoschi, Natalia Vélez-Ríos, and Jonathan P. Williams. Modeling Count-Valued Time Series with Discrete and Continuous Models *In Preparations*, 2026+.

    - Investigates the tradeoffs associated with modeling count-valued time series data using discrete models (e.g., INARMA) over classical continuous models (e.g., ARMA), and develops a software package to fit INAR models for common innovation distributions.

- Jason A. Osborne, Melody Wen, and **Neil Dey**. MLBDecideR: A Shiny App for Baseball. *Notices of the American Mathematical Society*, October 2020.
    - Introduces a web application that gives information/predictions regarding various in-game strategies in Major League Baseball by carrying out appropriate Markov chain calculations.

Industry Experience

Google | L4 Engineer *Kirkland, WA; July 2025 – Present*

- Frontend engineer for App Lifecycle Manager.
- Note: Hired under AI/Machine Learning interview track; prediction speciality
- Implemented background auto-refresh service, emergency rollback service, and easy integration with App Design Center
- Technology Stack: Angular, GraphQL, Java

Amazon | Applied Science Intern *Irvine, CA; June 2021 – August 2021*

- Designed and implemented a hierarchical Bayesian model to predict media selection behavior of individual Amazon customers
- Final model performed over 30% more accurately than existing model for behavior prediction
- Technology stack: MySQL, TensorFlow 2 for Python 3.

Boeing | Data Science Intern *Seattle, WA; June 2020 – August 2020*

- Implemented a Faster R-CNN computer vision model to track assembly line progress in factories
- Final model performed comparably to human employees
- Technology Stack: Tensorflow 1 for Python 3 (Object Dection API)

Amazon | Software Engineering Intern *Seattle, WA; June 2019 – August 2019*

- Architected and implemented a service to manage AWS accounts used for integration testing
- Created APIs for integration tests to automatically borrow and return accounts, preventing conflicts between different teams' automated tests
- Technology Stack: DynamoDB, AWS Lambda and Cloudwatch, Java

Cengage/WebAssign | Software Engineering Intern *Raleigh, NC; June 2018 – August 2018*

- Worked as a full-stack web developer, creating a single page web application tracking metrics regarding the WebAssign platform
- Views included a heatmap showing current users on the platform, a risk assessment tool determining when it is safe to deploy, and a visualization of current HTTP errors and response times
- Technology Stack: React, Node, LESS; Java, Spring-Boot, JDBC (MySQL), Dynatrace; Jenkins, JUnit, and Enzyme

Programming Language Proficiencies

- Highly Proficient: Python 3, Java
- Proficient: C, JavaScript, SQL
- Familiar: C#, R, Julia, Scala, x86 Assembly, SAS